

Failure mode measured on the VCS	VCS measured evidence proven in this benchmark	Neural-network analogue documented, not proven here	Status here → there
1 Gradient vanishes on discrete / index outputs	0.234 gradient bucket F (position) single record The NAIVE position gradient is provably zero at a strobe-timed argmax readout (\$3.2 oracle). The whole gradient/ correlational bucket stays low = 0.234 (n=9) on the position regime; even the best gradient method does not clear it.	Shattered / saturated gradients; non-differentiable argmax & index readouts (analogue refs G1, G2).	proven here ↓ analogue there
2 Saliency is model-invariant (sanity-check fails)	0.000 sanity-check margin single record Program-randomization test: saliency cosine = 1.000 to the real map while 96% of RAM changed -> sanity check FAILS. Constant Jacobian on the content path, no ReLU.	Saliency unchanged under model- & label-randomization; behaves like an edge detector (analogue ref S1).	proven here ↓ analogue there
3 Correlation ≠ causation (Granger & tuning trap)	0.136 Granger F 95% CI [0.09, 0.18] Granger-edge faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.213 (spurious) -- vs the exact \$3.2 oracle.	Spurious Granger edges; 'present ≠ used' tuning & attention-as-explanation (analogue refs C1, C2).	proven here ↓ analogue there
4 Single-unit ablation has an interaction blind-spot	0.207 connectome F 95% CI [0.13, 0.29] Single-unit lesion importance is faithful (0.965) yet MISSES wiring: the connectome graph is recovered at only 0.207; SAE ablations move y in 0 of the audited cells.	Single-unit ablation misleading; computation distributed across units (analogue refs D1, D2).	proven here ↓ analogue there
5 SAE/probe: present-not-used	0.108 selectivity F 95% CI [0.08, 0.14] SAEs match a variable (100% of candidates in 5/6 games, min 82%) yet score 0.173 on causal use; probes decode at 0.951 acc but 14 cells are decodable-not-causal -> selectivity F = 0.108.	Control-task selectivity gap; superposition / polysemantic neurons (analogue refs P1, P2).	proven here ↓ analogue there
6 IG/EG baseline sensitivity	0.175 EG F (all-regime) 95% CI [0.11, 0.24] IG magnitude is baseline-sensitive (corr. sens. up to 0.999 across games); a baseline = the content byte kills IG entirely; Expected Gradients lands at 0.175.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input (analogue refs B1, B2).	proven here ↓ analogue there

Analogue references (documented in the literature, not measured here):
 G1 Balduzzi et al. 2017; G2 Sundararajan et al. 2017 (IG). S1 Adebayo et al. 2018. C1 Anand et al. 2019; C2 Jain & Wallace 2019.
 D1 Morcos et al. 2018; D2 Leavitt & Morcos 2020. P1 Hewitt & Liang 2019; P2 Elhage et al. 2022. B1 Kindermans et al. 2019; B2 Sturmfels et al. 2020.